Journal of Building Engineering 98 (2024) 111117



Contents lists available at ScienceDirect

Journal of Building Engineering

journal homepage: www.elsevier.com/locate/jobe





# Automated evaluation of degradation in stone heritage structures utilizing deep vision in synthetic and real-time environments

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ARTICLE INFO

*Keywords:*
Deep learning
Computer vision
Heritage structure
Health monitoring
Synthetic environment
Convolutional neural networks

ABSTRACT

Preserving historical structures stands as a crucial pillar in the realm of sustainable development, where safeguarding our heritage takes precedence. However, the manual monitoring of these architectural marvels as they weather the passage of time remains a tedious and time-consuming endeavor. In this pursuit, computer vision emerges as a transformative tool, poised to alleviate or even eradicate the necessity for extensive human involvement in on-site assessments. The novel contributions of this study to the realm of knowledge encompass two key aspects: firstly, the utilization of synthetic images for categorizing vegetation, and secondly, the adaptation of the model for automated defect identification, culminating in the creation of an algorithm tailored for flaw detection in historical stone buildings. Present research endeavors to harness the prowess of the Mask Region-based Convolutional Neural Network (R-CNN) algorithm, a cutting-edge computer vision methodology, to discern, pinpoint, and delineate deteriorations within historical stone structures, classifying anomalies akin to vegetative growth. The model's training phase involves meticulously annotating anomalies within 501 synthetic images generated through Blender 3D software. Subsequently, the model undergoes rigorous testing, employing 428 photographs sourced from iconic UNESCO World Heritage sites across India, including Udayagiri and Khandagiri Caves, Golconda Fort, and Lingaraj Temple. The dataset expands to encompass a total of 2846 images, partitioned into an 80:10:10 ratio for training, validation, and testing, respectively. The Mask R-CNN-based ResNet101 model yields an impressive accuracy rate of 94 %, coupled with an Intersection over Union (IOU) of 94 %. Meanwhile, the Mask R-CNN-based ResNet50 model achieves a commendable accuracy of 93 %, with an IOU of 91 %. To gauge the efficacy of the proposed model, a comparative analysis is conducted with the state-of-the-art YOLOv8 model. Henceforth, the proposed methodology can be utilized to create an automated system for the visual inspection of stone-built heritage structures, to preserve them for future generations.

## 1. Introduction

UNESCO World Heritage structures stand as cultural and historical landmarks, encapsulating a nation's rich heritage and acting as a vital link connecting the past with the present. These architectural treasures often employ traditional building materials and methods, rendering them susceptible to damage from natural disasters or human activities. Preserving these heritage structures holds

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https://doi.org/10.1016/j.jobe.2024.111117
Received 4 November 2023; Received in revised form 22 April 2024; Accepted 19 October 2024
Available online 20 October 2024
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**Abbreviations**

UNESCO - United Nations Educational, Scientific and Cultural Organization
CNN - Convolutional Neural Network
FCNN - Fully Convolutional Neural Network
U-Net - U-shaped encoder-decoder network
VGG - Visual Geometry Group
ResNet - Residual Network
IOU - Intersection over Union
ROI - Region of Interest
LGBM - Light Gradient Boosting Machine
R-CNN - Region-based Convolutional Neural Network
RPN - Regional Proposal Network
RGB - Red Green Blue
GPU - Graphics Processing Unit
YOLO - You Only Look Once

immense importance as they mirror a nation's cultural essence and historical significance. Frequently, these edifices are enveloped by surrounding vegetation, which can potentially compromise their structural integrity. Consequently, it becomes imperative to monitor and analyze the growth of vegetation and its impact on these heritage structures to ensure their continued preservation. Remote sensing techniques, such as satellite and airborne imagery, have been extensively employed to monitor vegetation growth and its influence on historical structures [1,2]. With an expanding array of invaluable cultural assets to oversee, the potential for oversight errors amplifies, risking irreversible harm to these cherished relics. Hence, the imperative for pioneering inspection techniques is underscored. Efficiently conserving and managing heritage structures necessitates an intricate comprehension of their integral constituents, which encompass vegetation. Recent times have witnessed the extensive adoption of deep learning techniques in the domain of image segmentation, particularly for tasks involving the delineation of vegetation.

In the realm of image segmentation tasks, deep learning techniques, particularly CNNs, have exhibited considerable promise [3]. CNNs belong to a category of neural networks that have brought about a revolution in the field of computer vision, consistently achieving state-of-the-art performance across a range of image-processing tasks [4–6]. Notably, recent years have witnessed a surge in research endeavors exploring the application of CNNs for the multiclass segmentation of both concrete and heritage structures [7–10]. These investigations have employed diverse CNN architectures and optimization methods, ultimately yielding precise and reliable segmentation outcomes. The adoption of computer vision techniques, particularly the prevalence of CNNs, has surged in popularity in recent years [11–13]. The proliferation of vegetation on heritage structures can give rise to a spectrum of issues, encompassing structural degradation, corrosion, and decay. Consequently, the precision of vegetation segmentation emerges as a critical component in formulating apt conservation strategies tailored to heritage structures [14,15]. Segmenting vegetation on historical structures presents a formidable challenge due to the intricate nature of the flora and the pronounced diversity among the structures themselves. In this contemporary landscape, CNNs have risen as a potent tool for image processing tasks, including segmentation [16].

The CNN acquires the ability to discern pertinent features within the images, allowing it to discriminate between undamaged and damaged heritage structures. Following successful training, the CNN is then deployed in the testing phase to classify new images of heritage structures, categorizing them as either undamaged or damaged [17,18]. One of the pioneering pixel-based methods for detecting damage in heritage structures employed a CNN to categorize each pixel within an image of a bridge deck as either healthy or damaged [19]. This CNN underwent training using a dataset comprising 5000 images, yielding an accuracy rate of 90 %. Another noteworthy pixel-based approach employed a CNN to classify individual pixels in an image of a concrete structure as either healthy or damaged. The training dataset encompassed 1980 images, and this method achieved an accuracy of 93.8 % [20]. However, these methods may not be suitable for complex images with multiple objects, and they may not be able to capture the context of the damage. In object-based approaches, the CNN is trained to categorize each ROI as either an undamaged or damaged part in heritage structures. The training dataset encompassed 1500 images, and the method achieved an accuracy rate of 94 % [21]. Another noteworthy object-based technique utilized a CNN to differentiate ROIs in an image of a brick structure as either healthy or damaged, utilizing a dataset consisting of 1000 images, with an accuracy of 92 % [22]. However, it's worth noting that these methods may entail higher computational expenses and necessitate more memory resources compared to pixel-based techniques.

Yang et al. [23] introduced a CNN-based approach designed for the precise segmentation of vegetation surrounding heritage structures. The actual segmentation of vegetation was performed employing a U-Net segmentation model, as proposed by Ronneberger et al. [24]. Remarkably, this innovative methodology achieved an impressive accuracy rate of 95.3 % when tested on a dataset comprising 100 images of heritage structures adorned with vegetation. A multi-scale CNN-based technique was introduced for the segmentation of vegetation surrounding heritage structures [25]. Later, a method for the segmentation of vegetation from high-resolution images of heritage structures was proposed, employing a CNN-based methodology. Their approach incorporated a modified U-Net architecture, complete with skip connections and residual blocks, aimed at enhancing segmentation performance [26]. Liu et al. [27] introduced a novel technique for the segmentation of vegetation within multi-source remote sensing images depicting heritage structures, employing a CNN-based approach. Their method ingeniously combined the capabilities of a ResNet-101 model, as

introduced by He et al. [28], with a U-Net model. This fusion allowed the model to harness features from both high-resolution aerial images and low-resolution satellite images, enabling a comprehensive understanding of the data.

Zhao et al. [29] employed an innovative technique for the segmentation of vegetation within ground-based images depicting heritage structures, harnessing the capabilities of a CNN-based approach. Their method leveraged a U-Net model, augmented with a spatial pyramid pooling module designed to capture contextual information across multiple scales. Impressively, their proposed approach attained a notable accuracy rate of 95.2 % when assessed on a dataset comprising 200 ground-based images featuring heritage structures adorned with vegetation. It's worth noting that only a limited number of studies have ventured into both object detection and segmentation for heritage structures [30–32]. Another study delved into bridge crack recognition and quantification through hybrid feature learning, resulting in effective segmentation [33]. To assess the performance of neural networks such as VGG16, VGG19, ResNet-50, and Inception-v3 in crack detection and localization, a comparative analysis was conducted [34,35]. Additionally, for the classification of multiclass damages including spalling of façade, spalling of concrete, damage with exposed rebars, and severely buckled rebars (gathered from a post-disaster reconnaissance in Taiwan), deep learning techniques like Inception-v2, ResNet-50, ResNet-101, and Inception-ResNet-v2 were employed for predictive purposes [36].

The study focused on examining tunnels with defects, particularly cracks, using deep learning techniques like FCNNs. A comparative analysis was conducted between FCNN networks and the U-Net architecture for crack detection and segmentation [37]. In the realm of masonry structures, pattern recognition for damage detection was achieved through the utilization of classifiers such as support vector machines and neural networks. Leveraging a dataset comprising 1000 images, an impressive precision rate of 96.5 % was attained [38,39]. Various parameters including precision, accuracy, F1 score, IOU, sensitivity, and the dice coefficient were taken into account to evaluate the performance of different models or architectures [40]. Larger datasets are generally advantageous for effectively training deep learning models. In this regard, a dataset consisting of 40,000 images, with 20,000 images depicting cracked and non-cracked concrete surfaces each, was employed. This dataset yielded an impressive accuracy rate of 98 % when applied to different pixels [41–44]. Detecting damages within a dataset comprising 2000 images was notably influenced by texture. Challenges related to texture encompassed texture classification, synthesis, segmentation, and understanding the shape of the texture [44,45].

A two-stage feature generation approach was implemented, incorporating pixel values and geometric shapes, utilizing photographs of concrete structures, including those affected by shadows and dirt. LGBM was assessed, yielding remarkable outcomes with a 99.7 % accuracy, 75.71 % sensitivity, 99.9 % specificity, 68.2 % precision, and an F-measure of 0.6952 [46]. Furthermore, advancements have been made in various aspects such as classification, pixel-level concrete crack detection, and crack measurement [47]. Real-time inspection and damage detection in concrete structures have been achieved using CNNs [48,49]. Additionally, endeavors have been made toward concrete crack detection through handwritten scripts, employing R-CNN [50]. The limited focus on studies pertaining to stone structures can be attributed to several factors. Visual inspection of periodic stone monument degradation proves to be labor-intensive, especially daunting for large-scale immovable cultural heritage sites, where constraints in manpower and heightened error rates pose significant challenges. Misinterpretation of monument degradation may result in irreversible structural damage, necessitating alternative testing methodologies. Computer vision technologies present a promising avenue for mitigating field errors and overcoming associated challenges [51]. Patil et al. [52] conducted research involving the development of an open-source web application tailored for fault detection in heritage structures through an AI deep-learning model.

This paper explores the application of CNNs for segmenting vegetation on UNESCO World Heritage historical structures. Several significant challenges impede vegetation segmentation for historical structures. Firstly, there is a noticeable scarcity of annotated datasets tailored to this specific task. Existing datasets are often small and cover only a limited range of structures and vegetation types. Therefore, there is a pressing need to gather more diverse and comprehensive datasets to effectively train and evaluate CNN-based methods for vegetation segmentation on historical structures. Secondly, the inherent variability in historical structures and vegetation types poses a significant challenge. Historical structures exhibit differences in shape, texture, and construction materials, all of which can potentially impact the accuracy of vegetation segmentation.

Similarly, vegetation showcases a variety of shapes, colors, and densities, making it difficult to distinguish from the background. Thus, it is crucial to develop CNN-based techniques capable of accommodating this variability and demonstrating robust performance across various historical structures and vegetation types. Another critical challenge is the demand for efficient and real-time segmentation methodologies. Many existing CNN-based approaches designed for vegetation segmentation on historical structures require substantial computational resources and involve lengthy processing times. Consequently, there is an urgent need to develop lightweight and efficient CNN architectures capable of enabling real-time vegetation segmentation on low-power devices. In this paper, an automated approach is proposed for evaluating degradation in stone heritage structures utilizing deep vision techniques. The proposed method integrates synthetic and real-time environments to train and test deep learning models effectively. Synthetic images are generated to create a diverse and comprehensive dataset, addressing the limitations of annotated datasets. The deep vision system is trained to detect various forms of degradation, including vegetation, cracks, efflorescence, and missing components. Real-time evaluation is facilitated through lightweight and efficient deep learning architectures, enabling rapid and accurate assessment of degradation in stone heritage structures.

## 2. Proposed model for the deterioration detection

The primary objective of this study is to create a fully automated visual inspection system that can replace existing, expensive methods, particularly when dealing with limited dataset sizes. Furthermore, the aim is to ensure that conducting such research is not overly complex, making it accessible to individuals familiar with structural health monitoring practices. In the realm of computer vision, Mask R-CNN stands out as a widely acclaimed deep-learning framework for addressing the challenge of semantic segmentation

[53]. The present investigation revealed the Mask R-CNN model exhibited exceptional performance in object identification and masking when compared to previously available frameworks. Therefore, the Mask R-CNN model is adopted as the foundation for a new application framework designed for defect detection and localization. The integral components of a Mask R-CNN model encompass an RPN, an ROI alignment layer (ROI Align), a Bounding-Box Object Detection Head, and a Mask Generation Head, as illustrated in Fig. 1. The Faster R-CNN algorithm incorporates the first four components, with the RPN serving to propose candidate bounding boxes for potential objects.

## 3. Backbone and RPN

The Mask R-CNN model was employed to produce output that includes classification along with bounding boxes and masks to delineate defects. In the process, Adam's algorithm was used, which employs stochastic gradient descent to calculate an appropriate learning rate for each weight parameter. This dynamic adjustment ensures that weights frequently updated receive smaller learning rates, while those updated less frequently receive larger ones. This feature makes Adam's algorithm a valuable optimizer for model training due to its adaptive learning rate technique. ResNet is commonly structured as a sequence of convolutional layers coupled with residual connections. The architecture comprises multiple residual blocks, each housing several layers interlinked by residual connections. Notably, as the spatial resolution of the feature maps diminishes, the convolutional layers within these blocks exhibit an escalating number of filters. RPN operates by examining features extracted from the input image, producing a collection of region proposals that serve as potential object bounding boxes. At each position within the feature map, the RPN generates a set of anchor boxes. Each anchor box represents a prospective object proposal and is defined by its central coordinates and aspect ratio.

For each anchor box, the RPN predicts two critical values: firstly, the probability denoting the likelihood of the box containing an object, and secondly, the adjustment in position concerning the default anchor box. These predictions enable the selection of the most promising anchor boxes, and the positional adjustments refine the box locations. Subsequently, a process known as non-maximum suppression is applied to eliminate redundant proposals, culminating in the generation of a finalized set of object proposals. RPN undergoes end-to-end training alongside the remaining components of the Mask R-CNN architecture. This training employs a multi-task loss function that combines the losses for object-ness classification and bounding box regression. Consequently, the RPN learns to produce top-tier object proposals that closely align with the actual object bounding boxes. In summary, the RPN serves as a potent instrument for generating superior object proposals, thus contributing to the Mask R-CNN's ability to attain state-of-the-art performance across diverse object detection and instance segmentation assignments.

## 4. Anchor boxes

Mask R-CNN utilizes anchor boxes to facilitate the detection of multiple objects, including those of various sizes and objects that may overlap within a given image. This approach results in swifter and more precise object detection. Anchor boxes represent

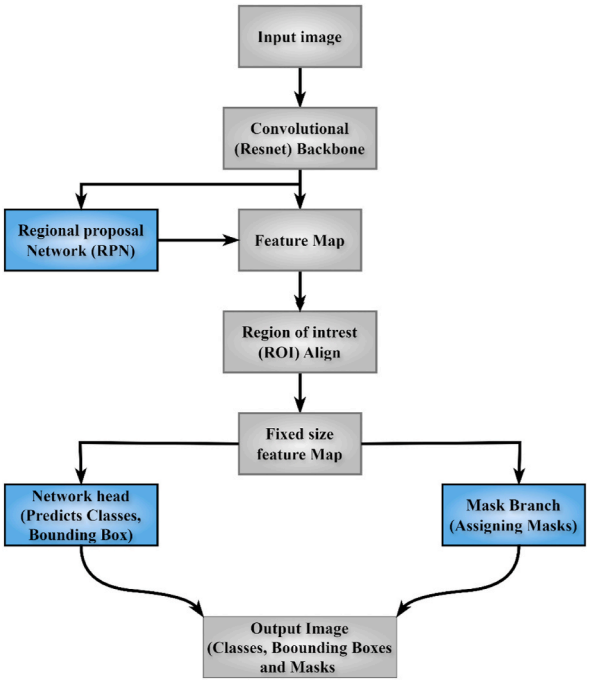



**Fig. 1.** Modified Mask R-CNN-based Res-Net Backbone network.

predefined rectangular bounding boxes with fixed dimensions, offering the flexibility to tailor their size and shape to effectively match the objects of interest. To predict multiple objects or numerous instances within an image, Mask R-CNN generates a multitude of predictions. In the final stage of object identification, anchor boxes associated with the background class are removed, and the remaining ones are filtered based on their confidence scores. The selection process employs an IOU threshold of 0.5 to identify suitable anchor boxes. Ultimately, the highest-confidence anchor boxes are chosen through the application of Non-Maximum Suppression.

## 5. ROI pooling

A crucial element of Mask R-CNN is the ROI pooling layer, meticulously crafted to extract a consistent-sized feature map for each proposed region. This layer receives the feature map generated by the backbone network along with a set of region proposals defined by bounding boxes described in terms of coordinates (*x, y, w,* and *h*). Here, (*x, y*) corresponds to the top-left corner coordinates of the proposal, while (*w, h*) denotes its width and height, respectively. The ROI pooling layer then partitions each region proposal into a fixed number of equally-sized rectangular bins, regardless of the actual dimensions of the proposal. Within each bin, the ROI pooling layer employs max pooling to produce a standardized feature map for each proposal. Subsequently, the resultant fixed-sized feature map for each proposal proceeds through subsequent interconnected layers, all aimed at object discovery and instance segmentation. By utilizing the ROI pooling layer to generate consistent-sized feature maps, Mask R-CNN adeptly manages region proposals of diverse sizes and aspect ratios, ultimately ensuring precise object masks. The ROI pooling layer represents a pivotal component of Mask R-CNN's architecture, enabling it to attain cutting-edge performance across a broad spectrum of object recognition and instance segmentation tasks. The bounding box coordinates are then used to change the size of the mask and line it up with the original image.

## 6. Loss functions

The Faster R-CNN function encompasses both classification loss and bounding box loss, while Mask R-CNN, which also performs instance segmentation, includes instance segmentation loss in its function. Notably, instance segmentation loss exclusively considers positive cases, where the IOU value exceeds zero. The comprehensive loss function for Mask R-CNN can be expressed as follows:

$$Total\ Loss = L_{cls} + L_{box} + L_{mask} \tag{1}$$

where $L_{cls}$ is classification class, $L_{box}$ is a box (regression) loss and $L_{mask}$ is mask loss. The classification loss measures the accuracy of the object classification task. It is computed using the binary cross-entropy loss function over the predicted class probabilities

$$L_{cls} = -\sum i[y_i * log(p_i) + (1 - y_i) * log(1 - p_i)] \tag{2}$$

where $y_i$ is the ground-truth label for the $i$ th proposal, and $p_i$ is the predicted probability of the proposal being a positive example. But in the architecture, only one object detection is carried out, so

$$L_{cls} = -\sum log(p_i) \tag{3}$$

The bounding box regression loss measures the accuracy of the predicted object bounding boxes. It is computed using the smooth $L1$ loss function as,

$$L_{box} = \sum_{i=1}^{n} (smooth_{L1}(t_i - v_i)) \tag{5}$$

$$smooth_{L1}(X) = \begin{cases} 0.5x^2, |x| < 1 \\ |x| - 0.5, otherwise \end{cases} \tag{6}$$

where $t_i$ is the predicted bounding box coordinates, $v_i$ is the ground-truth bounding box coordinates, and $n$ is an indicator variable that is equal to 1 if the $i$ th proposal is a positive example and 0 otherwise. The mask segmentation loss measures the accuracy of the predicted object masks. It is computed using the binary cross-entropy loss function over the predicted mask pixels:

$$L_{mask} = -\left(\frac{1}{N}\right) \sum i * log(mask_{pred_i}) + \left(\frac{1}{N}\right) \sum i * log(1 - mask_i) \tag{7}$$

where $N$ is the total number of mask pixels, $mask_i$ is the ground-truth mask for the $i$ th pixel, and $mask_{pred_i}$ is the predicted mask for the $i$ th pixel.

The Mask R-CNN's mask branch is trained using a binary cross-entropy loss function, which penalizes the model for incorrect pixel-wise predictions. This loss function is specifically computed for positive proposals, i.e., those that contain an object. The training process involves a combination of various loss functions, including binary cross-entropy, classification, and bounding box regression losses. A key advantage of Mask R-CNN is its capability to handle objects of diverse sizes and shapes effectively. This is achieved through the use of the ROI pooling layer, which converts each proposal into a fixed-size feature map. This ensures that the model can provide accurate predictions for objects with varying sizes and shapes. Additionally, Mask R-CNN's ability to perform instance segmentation, encompassing both object detection and pixel-level segmentation, is another notable feature of this architecture.



(a) Golconda Fort (Hyderabad) and Lingaraj temple (Bhubaneswar), India



(b) Images having defects collected from stone heritage structures

**Fig. 2.** UNESCO World Heritage Stone structures.

## 7. Experimental investigation

The dataset used in this study was collected from three prominent UNESCO World Heritage historical sites: the Lingaraj temple, Udayagiri, and Khandagiri caves in Bhubaneswar, as well as the Golconda site in Hyderabad, India as shown in Fig. 2 (a). This dataset encompasses a range of observed defects, including vegetation, cracks, missing parts, delamination, and efflorescence. However, the primary focus of this study centers on the vegetation class, given its prevalence as the most common form of damage affecting historical stone structures. Fig. 2 (b) illustrates the images having defects collected from stone heritage structures. The investigation primarily concentrates on generating synthetic images related to vegetation. This approach involves the utilization of synthetic environments created using Blender 3D software. The synthetic images are primarily oriented toward simulating and analyzing the growth of vegetation, particularly in instances where it has led to damage to the temple walls (see Fig. 3).

## 8. Data collection: real and synthetic environment

A total of 428 real-world images are collected from UNESCO World Heritage historic stone structures i.e., Lingaraj temple, Udayagiri, Khandagiri caves in Bhubaneswar, Odisha, and Golconda Fort in Hyderabad, India. However, it's worth noting that the dataset size in these studies was relatively small. To address this challenge, a synthetic environment-based approach was adopted, involving the generation of synthetic data for model training [53,54]. This study underscores the utility of synthetic images in visual inspection tasks, particularly when dealing with limited dataset sizes. In this context, Blender 3D software is utilized to generate synthetic images. Blender stores information in several different "data blocks" or file types, including objects, meshes, lights, scenes, materials, images, and so on. Blender objects are composed of numerous data blocks that are connected to form the final product.

Blend files may be used as reusable resource libraries since objects, models, materials, textures, and so on can be connected to other blend files. The color, size, and density of the synthetically generated images are made in such a way that a lot of variations between all images. These will help the dataset cover all the variations. The wall background (surface) needs to be generated in such a way that it should be similar to that of real-world images. These synthetic images were incorporated into the dataset and used for model training, subsequently tested on real-world images. The synthetic images generated from the Blender 3D, a total of 501 synthetic images are generated using the synthetic environment, and the real-world captured images are added to the dataset. The synthetically generated images and the real-world images are taken into the Roboflow software for annotating and labeling. The ground truth label consists of a

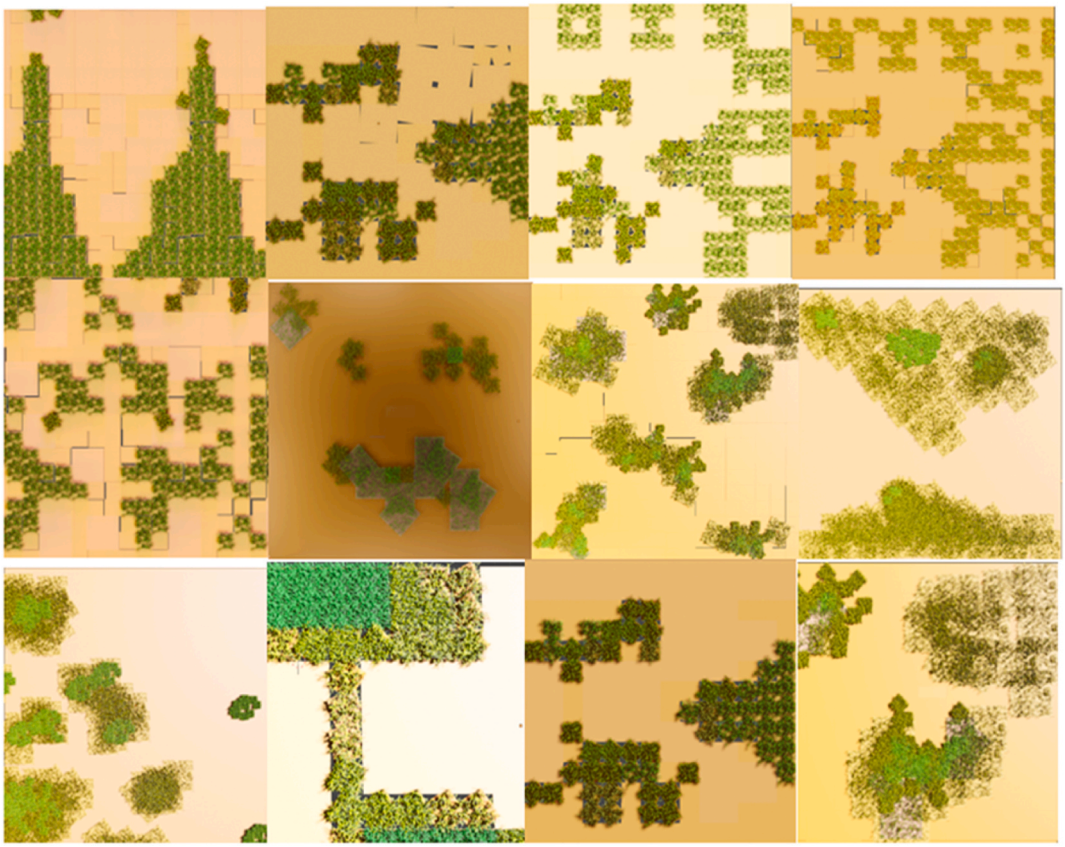


**Fig. 3.** Synthetic images generated from Blender 3D software.

background and grass. The background of the real world and synthetic images need to be corrected so that the training model will give good results. The images need to be created in such a way that there should be a separation between the clusters of the vegetation.

## 9. Data pre-processing

The initial and paramount phase in machine learning model development is known as data pre-processing, which involves converting raw data into a format suitable for machine learning. In real-world scenarios, data is often neither pristine nor well-structured, making this stage crucial. Notably, in the dataset used for this study, the photos were already pre-cropped, obviating the need for this particular pre-processing step. In addition to the cropping step, the pre-processing phase encompassed the conversion of RGB photos into monochrome images and the overall cleaning of the images. After the conversion of RGB images into monochrome, the ensuing steps in the pre-processing pipeline entailed image scaling, data annotation, and data augmentation, carried out sequentially.

## 10. Data annotation

Data annotation is a crucial process involving the identification and labeling of relevant features within images, which is particularly effective for supervised machine learning applications. For a system to analyze, understand, learn patterns, and generate appropriate outputs, it relies on labeled datasets as input. In this study, the images were annotated using Roboflow, a platform designed for Computer Vision developers that streamline data collection, pre-processing, and model training. Roboflow offers compatibility with a wide range of annotation file formats, facilitating tasks such as image rotations, resizing, contrast adjustments, and data augmentation as part of the data pre-processing phase. Teams can seamlessly manage the entire process within the platform, from data collection to annotation. Creating annotations to identify the location of objects within images is as straightforward as drawing bounding boxes around them and providing descriptive labels. In this particular context, manual labeling with ground truths was employed within the software to ensure precise annotations (see Fig. 4).

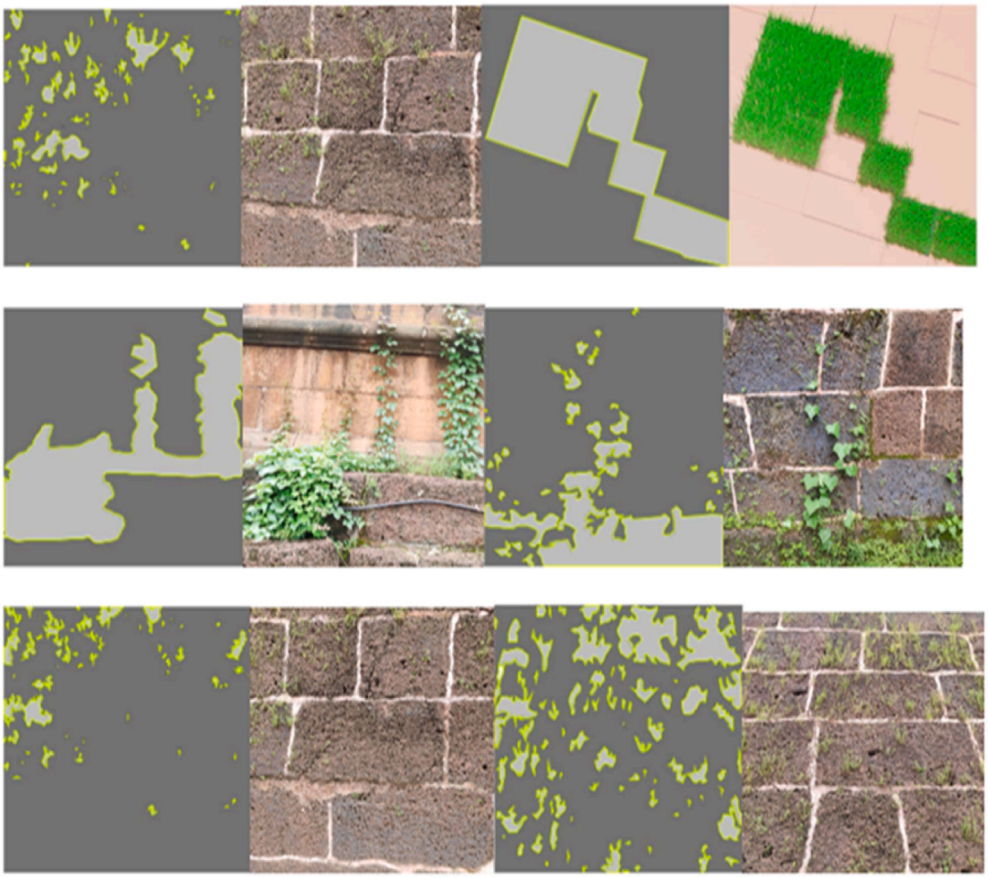


**Fig. 4.** Data annotation using Roboflow.

## 11. Data augmentation

Data augmentation is a widely used technique, particularly in cases where obtaining a sufficient quantity of high-quality data for training robust AI models can be challenging. The core of data augmentation methods typically revolves around position augmentation and color augmentation algorithms. It becomes particularly valuable when acquiring a substantial volume of data proves challenging. The capacity of a model to generalize effectively is often contingent on the diversity of the training data it receives, and image augmentation plays a pivotal role in achieving this diversity. Common data augmentation techniques include flipping, 90-degree rotations, random rotations, cropping, random shear, exposure adjustments (brightness changes), blurring, and noise addition, as illustrated in Fig. 5. A total of 501 images were generated using a synthetic environment.

To enhance the dataset for model training, these 501 images were utilized for augmentation, resulting in an expanded dataset comprising 1569 images. Additionally, 428 images were collected from nearby historical structures, and through augmentation techniques, the dataset size was increased to 1277 images. These images served as the training data for the model, while real-world images (collected images) were reserved for testing purposes. The generated synthetic images were employed exclusively for training the model. Subsequently, a combination of synthetic and real-world images was used for validation, and real-world images were set aside for model testing. The dataset, consisting of a total of 2846 images, was partitioned into a ratio of 80:10:10 for training, validation, and testing of the model, respectively. The testing set comprised 290 images, with an equal split between synthetic and real images, resulting in a 1:1 ratio, i.e., 145 synthetic and 145 real images. It's worth noting that this test dataset was not augmented, signifying that these images are entirely new to the model during testing. The dataset includes original images and their augmented counterparts, with corresponding sets for both real and synthetic images, as illustrated in Table 1.

The modified Mask R-CNN models based on ResNet50 and ResNet101 architectures were trained, validated, and tested using a dataset comprising 2846 images. The images were resized to a uniform size of 640x640 pixels, and the training process utilized a batch size of 64 and ran for 100 epochs. The choice of 100 epochs was determined through analysis and trial-and-error to strike a balance between minimizing the loss function and considering time and computational efficiency. Both labeled images for training and testing were included in the dataset. Fig. 6 (a) and (b) display visualizations of these images, showcasing how defects are segmented with different colors. This visualization aids in identifying the number of instances in each image. Real-world captured images typically contain more instances compared to synthetically generated images.

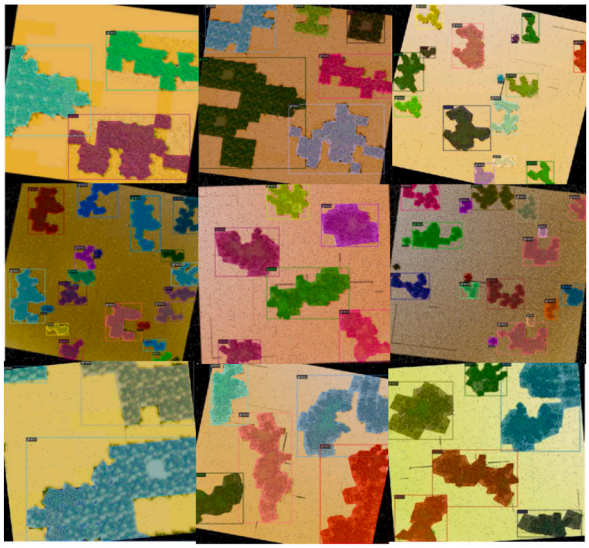
A total of 53,900 instances of grass as objects were observed across the 2846 images. Synthetic images with varying numbers of



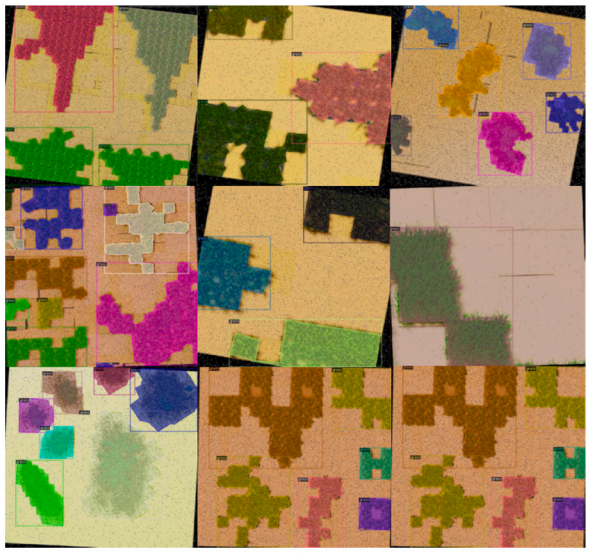
**Fig. 5.** Images generated using the Augmentation technique.

**Table 1**
The description of the dataset (images).

| Image Type | Original images | Augmented images | Total images |
|---|---|---|---|
| Real-world images | 428 | 849 | 1277 |
| Synthetic images | 501 | 1068 | 1569 |
| Total images | 929 | 1917 | 2846 |



(a) for Mask R-CNN-based ResNet50 model



(b) for Mask R-CNN-based ResNet101 model

**Fig. 6.** Visualization of training images.

object instances were included in the training dataset to reduce the model's sensitivity. The training dataset consisted of 2279 images, and the training process was capped at 10,000 iterations due to GPU limitations in Google Colaboratory. A learning rate of 0.005 was set to optimize model convergence. For testing the model's performance and defect prediction on images, a separate test dataset containing 290 images was employed. The modified Mask R-CNN model is trained using various combinations of datasets. It is observed that the dataset containing both synthetic and real-world images in the training dataset yields superior results. Consequently, a dataset comprising 1132 real-world images and 1424 synthetic images is utilized for model training. Additionally, 145 images from both the synthetic and captured image sets are separated for testing the model, with no augmentation applied to them. Several metrics

are considered for evaluating the performance of the Mask R-CNN models namely, average precision, IOU, and total losses. Both models are trained in the Google Colaboratory environment. Figs. 7 and 8 present the results for ResNet50 and ResNet101 architectures, respectively. The ResNet50 and ResNet101-based models, achieved the highest average precision values of 0.93 and 0.94, respectively.

The higher value achieved by the ResNet101 backbone Faster R-CNN model indicates its superior ability to predict multiple object classes with greater accuracy when compared to the ResNet50 model. Regarding loss, both models incorporate a combination of losses in class, box, and mask. The individual class, box, and mask loss values for the ResNet50 model are 0.15, 0.25, and 0.2, while the ResNet101 model's values are 0.1, 0.25, and 0.15, respectively. Regarding loss, the combination of all losses results in the total loss, which is 0.55 for the ResNet50 model and 0.45 for the ResNet101 model. As the number of iterations increases, the models achieve higher accuracy and minimal loss.

Furthermore, when the number of iterations is augmented, both models provide more precise masks for objects, resulting in reduced false negatives and false positives. One of the primary distinctions between the two models lies in their depth. The ResNet101 model boasts a depth of 101 layers, whereas the ResNet50 model comprises 50 layers. The ResNet101 model's increased depth enables it to capture more intricate and complex features and patterns within images. This is attributed to the fact that deeper neural networks possess the capacity to acquire more advanced data representations. To assess the model's performance, a series of real-world images and synthetic images that were not used during the model's training are utilized as input. The corresponding outputs for some of these images are presented in Figs. 9 and 10 for the ResNet50 and ResNet101-based Mask R-CNN models, respectively. Both ResNet models accurately predict the "grass" class, providing segmentation masks and bounding boxes with the highest confidence score of 100. The lowest confidence score achieved is 70 for the shallower ResNet50 model and 71 for the deeper ResNet101 model. Notably, testing on synthetic images yields confidence scores exceeding 95 for all images. This discrepancy can be attributed to the presence of fewer instances in synthetic images compared to real-world captured images. The accuracy approaching 95 % for the Mask R-CNN models indicates that all objects or instances within the images are successfully detected. Once the final dataset has been trained, the classifier is deployed to identify defects in photographs taken at the heritage site. The results are highly satisfactory, with detection confidences ranging from 71 % to 100 % for many issues. The detection confidence level is prominently displayed in the upper left corner of each

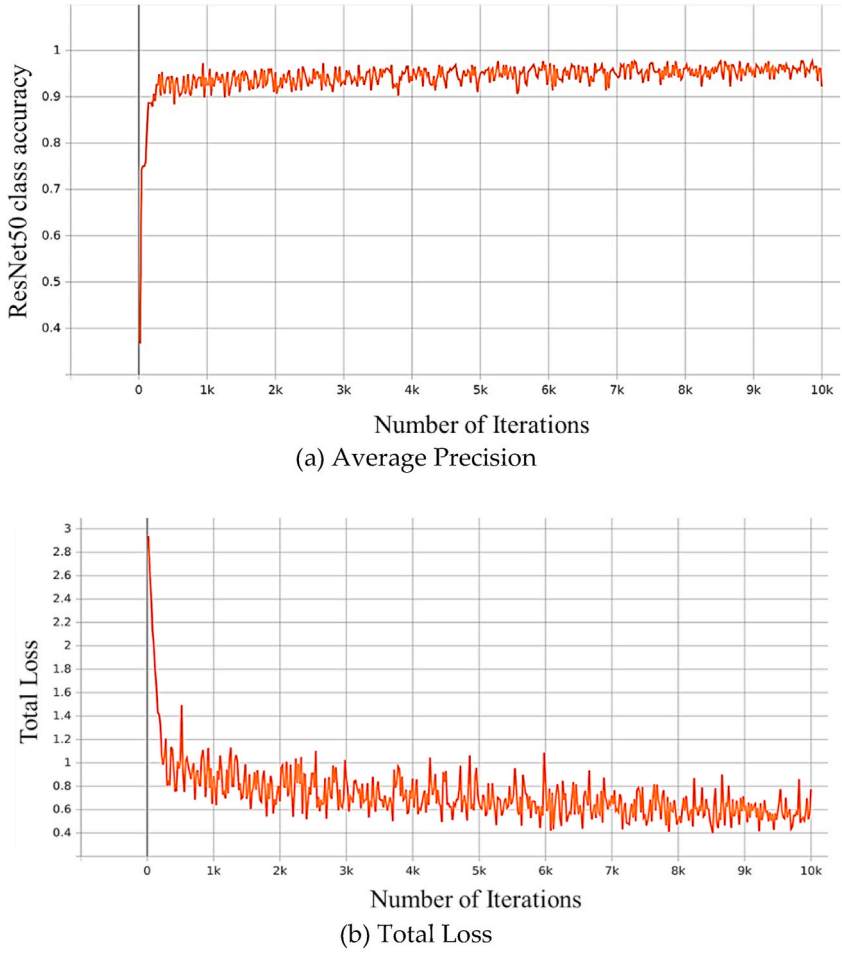



**Fig. 7.** Defect detection results indicating total loss for Mask R-CNN-based ResNet50 model.

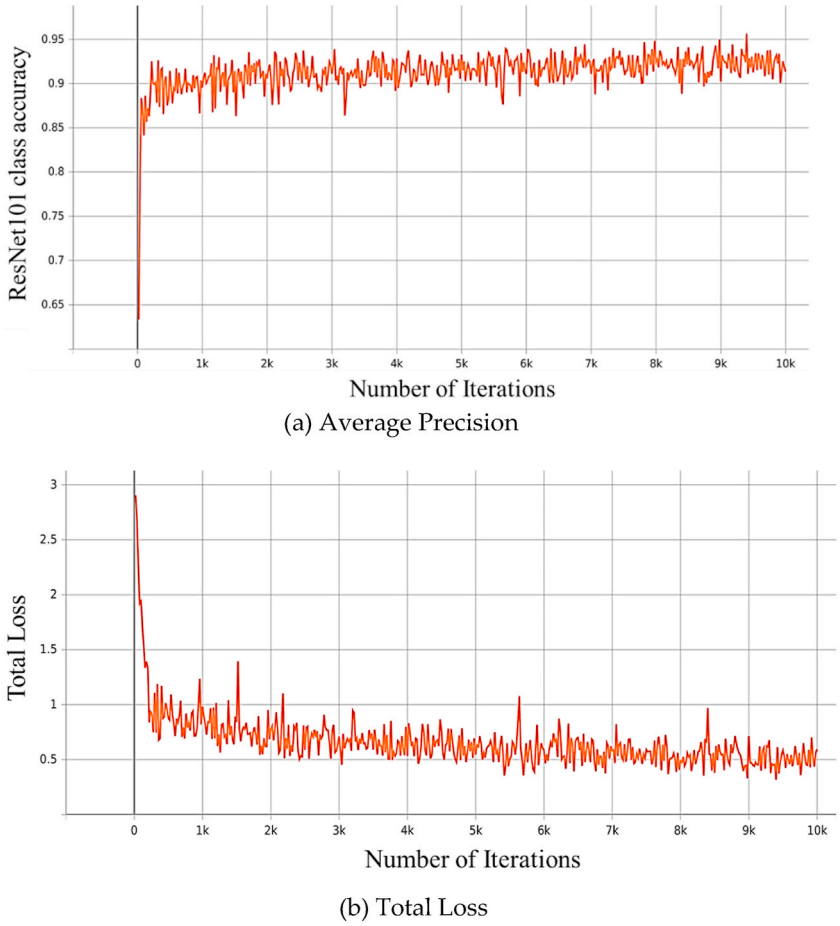



**Fig. 8.** Defect detection results indicating total loss for Mask R-CNN-based ResNet101 model.

identified bounding box. Various site conditions are considered to comprehensively detect defects. Notably, the model consistently excels in vegetation detection. The ResNet101-based Mask R-CNN model outperforms the ResNet50 model in terms of accuracy. This superiority arises from the ResNet101 model's capacity to capture intricate features and patterns in images, resulting in enhanced accuracy in such tasks.

Additionally, there is a divergence in computational complexity between the two models. The ResNet101 model incorporates more parameters and demands greater computational resources for training compared to the ResNet50 model. Consequently, training the ResNet101 model may necessitate more time and computational power, with the training duration being twice that of the ResNet50 model. Therefore, it is reasonable to adopt the ResNet50 model for efficiency, but for tasks where higher accuracy is paramount, the ResNet101 model is the preferred choice. Table 2 provides a comparison of the modified R-CNN-based Resnet models in terms of overall average precision, loss, and IOU. To assess and contrast the performance of the proposed modified Mask R-CNN model, based on the ResNet architecture, with the YOLOv8 network all three models were trained using the same dataset for vegetation detection. YOLOv8 is a real-time object detection model designed for speed and efficiency. They employ a single neural network to directly predict bounding boxes and class probabilities for multiple objects in an image, enabling them to achieve high frame rates and low latency.

However, the YOLO model may exhibit lower accuracy than some other object detection models, particularly when confronted with small objects or complex scenes. The mean Average Precision scores for the modified Mask R-CNN based on Resnet101 and YOLOv8 were 0.94 and 0.745, respectively, showcasing comparable detection accuracy. Notably, the proposed model delivered results in only half the time when compared to YOLOv8. The primary advantage of the Resnet architecture-based Mask R-CNN lies in its ability to detect defects with superior accuracy.

## 12. Conclusions

In recent years, advancements in deep vision techniques have provided promising solutions for the automated evaluation of degradation in heritage structures. Deep learning models, particularly CNNs, have demonstrated remarkable capabilities in image analysis and pattern recognition tasks. By leveraging deep vision techniques, it is possible to automate the process of degradation

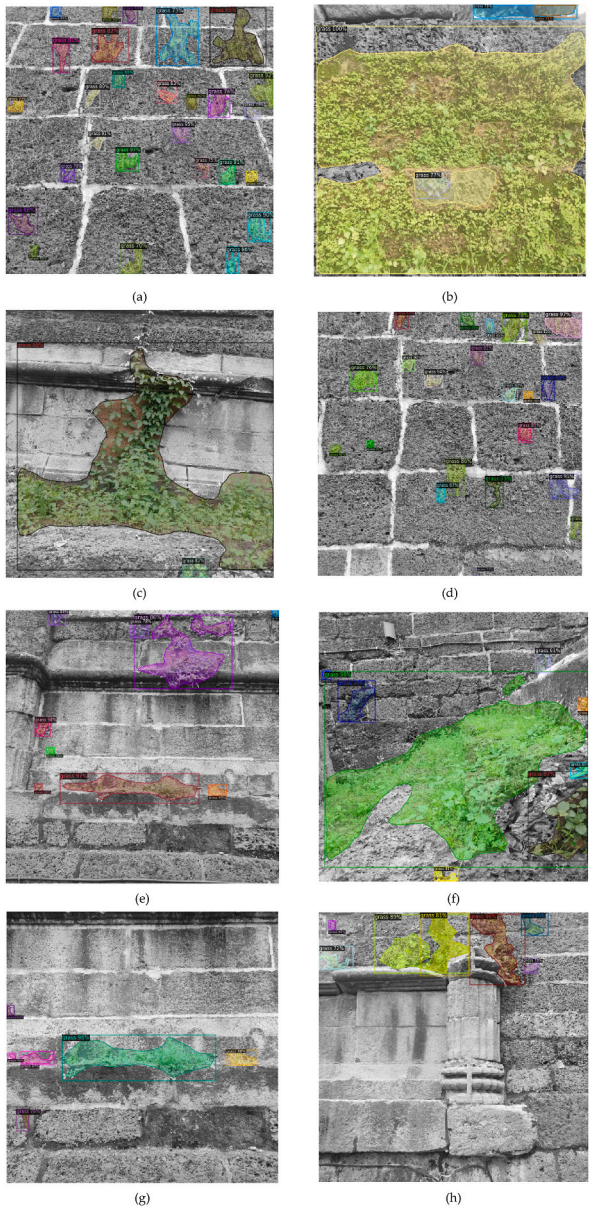



**Fig. 9.** Performance of Mask R-CNN ResNet50 on synthetic and real-time images.

evaluation, thereby improving efficiency and accuracy. This paper presents an automated approach leveraging deep vision techniques for the evaluation of degradation in stone heritage structures. The proposed method combines synthetic and real-time environments to train and test deep learning models effectively. By utilizing synthetic images, a diverse and comprehensive dataset is created, addressing the limitations of annotated datasets. The primary focus is on identifying defects in the vegetation category of historical buildings. The proposed approach leverages customized versions of the ResNet50 and ResNet101-based Mask R-CNN deep learning models to achieve automatic defect detection. The dataset used for model training comprises a total of 2846 images, including 1569 synthetic images and 1277 real-world images collected from Udayagiri, Khandagiri, Golconda Fort, and Lingaraj Temple. An important advantage of incorporating synthetic images into the dataset, along with the proposed model configurations, is the ability to detect defects in diverse backgrounds and from various angles, particularly when the original dataset has a limited image size. The ResNet50 and ResNet101-based Mask R-CNN models demonstrate the highest detection accuracy, with mean IOU values of 0.91 and 0.94, respectively. The total loss values were 0.6 and 0.5 for the ResNet50 and ResNet101 models, respectively.

The models were subjected to testing using real-world images and successfully identified all vegetation objects, achieving mean IOU values ranging from 0.71 to 1.0. The inclusion of synthetic images in this study expanded the dataset size, proving advantageous for model training. Further enhancement in the detection of various vegetation defects at heritage sites can be achieved by introducing

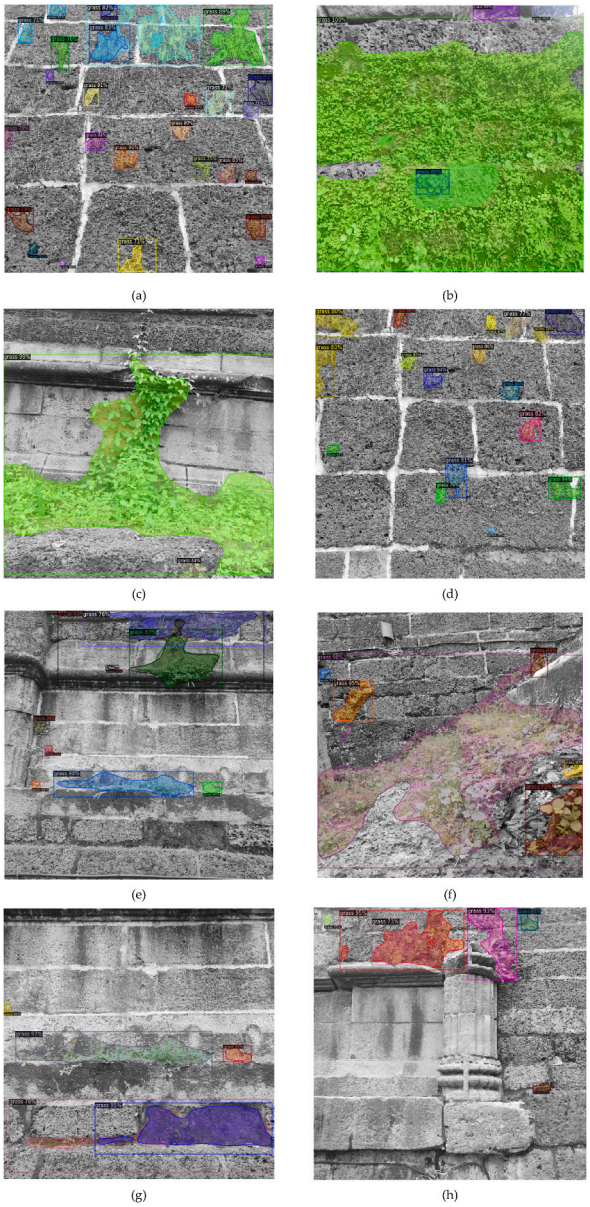



**Fig. 10.** Performance of Mask R-CNN ResNet101 on synthetic and real-time images.

**Table 2**
Comparison between Mask R-CNN training models.

| Model/Metrices | ResNet50 | ResNet101 |
|---|---|---|
| Mask R-CNN Average Precision | 0.93 | 0.94 |
| Mask R-CNN Total Loss | 0.6 | 0.5 |
| IOU | 0.91 | 0.94 |

additional images with diverse vegetation and background variations. Real-time evaluation is facilitated through lightweight and efficient deep learning architectures, enabling rapid and accurate assessment of degradation in stone heritage structures. This study's innovative contributions to the field of knowledge include the utilization of synthetic images for the vegetation category and the adaptation of the model for automated defect identification, resulting in the development of an algorithm for flaw detection in historical stone buildings. In this research, the proposed method is focused solely on detecting a specific type of issue commonly found in historic stone buildings, which is vegetation. Experimental results demonstrate the effectiveness of the proposed approach in

accurately identifying and evaluating degradation, paving the way for enhanced preservation efforts in cultural heritage conservation. Future research can focus on expanding the scope of degradation types beyond vegetation, cracks, efflorescence, and missing components. Investigating additional forms of degradation such as weathering, erosion, and structural deformities can provide a more comprehensive understanding of the condition of stone heritage structures. Incorporating multimodal data sources such as infrared imaging, LiDAR scanning, and thermal imaging can enrich the evaluation process by capturing different aspects of degradation not visible in RGB imagery alone. Integrating these data sources with deep vision techniques can enhance the accuracy and robustness of degradation detection models. Consequently, further research can focus on optimizing deep learning architectures to improve the efficiency and speed of real-time evaluation in low-power devices. Developing lightweight models and optimizing inference algorithms can enable rapid and seamless evaluation of degradation in stone heritage structures in real-time scenarios.

## CRediT authorship contribution statement

**T. Jothi Saravanan:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Visualization, Writing – original draft. **Bhukya Bhaskar:** Formal analysis, Investigation, Software, Validation, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgment

The corresponding author would like to thank the Virtual and Augmented Reality Center of Excellence (VARCoE) of the Indian Institute of Technology Bhubaneswar Research and Entrepreneurship Park for providing financial assistance for the work presented in this manuscript through the project code VARCoE/23/03. The second author would like to thank the Ministry of Education, Government of India, for the financial assistance provided during the research work.

## Data availability

No data was used for the research described in the article.

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